CS60017 Assignment

Opinion Maximisation on ego-Facebook (Abebe et al., KDD 2018)

[Deadline: 13.11.2026, 11:59 PM IST]

IMPORTANT INSTRUCTIONS

Plagiarism Alert

- Plagiarism of any kind is not allowed.
- If the similarity checker finds 60% or more code overlap between two submissions, both the SOURCE and the TARGET get a 100% penalty.
- Do not share your code or report with anyone, during or after the assignment.
- If a majority of the AI-text detectors we run flag a submission as AI-generated, it will be treated as 100% plagiarism.
- **Code error:** if your code does not run, marks are given for the logic alone, and you will be asked to explain it to a TA.
- **Library restrictions:** `numpy`, `scipy` (including `scipy.linalg`, `scipy.sparse.linalg` and `scipy.optimize`), `networkx`, `pandas`, `matplotlib` and the Python standard library are fine. `cvxpy` may be used only for the small-graph checks in Task 2. Packages that implement opinion dynamics or opinion optimisation are not allowed. The equilibrium computation, the greedy algorithm, the rank-one updates and the baselines must be your own code.

SUBMISSION INSTRUCTIONS

Submit the following files:

Assignment2_<ROLL_NO>_greedy_task1.py
Assignment2_<ROLL_NO>_paper_task2.py
Assignment2_<ROLL_NO>_groundtruth_task3.py
Assignment2_<ROLL_NO>_analysis_task4.py
Assignment2_<ROLL_NO>_bonus_task4.py
Assignment2_<ROLL_NO>_report.pdf

in a zipped folder named: **Assignment2_<ROLL_NO>.zip (or tar.gz)**

Your report.pdf must contain:

1. **[Mandatory]** Your **Roll Number** on the first line.
2. **[Mandatory]** Library requirements and the exact Python version used.
3. **[Mandatory]** Design details: how you compute the equilibrium, how innate opinions and resistances were generated, the values of ℓ, u and k you used, your random seeds, and the command that produces each result.
4. **[Mandatory]** Results and discussion for Tasks 1 to 4, with the required table and plots.

IMPORTANT: follow the file names and the per-task instructions exactly. Any deviation costs marks. See the **EVALUATION GUIDELINES** at the end.

Every person has an innate opinion and a resistance, which is how strongly they keep their own view instead of averaging their friends'. Instead of changing what people believe, this paper changes how easily they are persuaded. You must choose k people and reset their resistance so that the total opinion in the network goes up (or down) as much as possible. The objective is not submodular, so greedy has no guarantee here, and part of the job is to find out how well it does anyway.

Reference paper

Rediet Abebe, Jon Kleinberg, David Parkes, Charalampos E. Tsourakakis. Opinion Dynamics with Varying Susceptibility to Persuasion. KDD 2018, pp. 1089–1098.

Dataset

ego-Facebook, Facebook friendship circles combined into one graph: https://snap.stanford.edu/data/ego-Facebook.html

- 4,039 nodes and 88,234 undirected edges in a single connected component. Use `facebook_combined.txt`.
- The file lists each edge once. Build an undirected simple graph and check your counts against the numbers above. In the report, mention every preprocessing step.
- Report all target sets with the original SNAP IDs (This is to ease our checking, not adhering to this may cost marks).

Problem setting

- **Model–** Node i has an innate opinion s_i in [0,1] and a resistance α_i in (0,1]. Opinions evolve as $x_i(t+1) = \alpha_i s_i + (1 - \alpha_i) \cdot (1/d_i) \sum_{j \in N(i)} x_j(t)$. With A = diag(α) and P the random-walk matrix of the graph ($P_{ij} = 1/d_i$ for each edge), the equilibrium is $z = (I - (I - A)P)^{-1} A s$.
- **Objective–** $f(\alpha) = \sum_i z_i$, the total equilibrium opinion.
- **Interventions–** For a target set T with $|T| \leq k$, you may reset α_i to any value in [ℓ, u] for each i in T. Everyone else keeps their original α_i. Use ℓ = 0.001 and u = 1. *Maximisation* asks for the largest f and *minimisation* for the smallest. The *unbudgeted* version lets you change every node.
- **Parameters–** Draw the original resistances α_i uniformly from [0.001, 1]. Use two opinion profiles: (U) s_i uniform on [0,1], and (P) s_i drawn from Beta(0.5, 0.5).
- **Budget–** K = 30. Report results for every k among 1, 2, 3, 4, 5, 10, 20 and 30.

Task 1: Equilibrium and a baseline greedy

1. For two nodes a–b, derive z by hand as a function of s_a, s_b, α_a and α_b, and check your code against it for three choices of the parameters. Also confirm that iterating the update rule and solving the linear system agree on ego-Facebook to within 10^{-8}. A sparse direct solver is slow on this graph because of fill-in; an iterative solver such as `bicgstab` is much faster.
2. The paper proves that some optimal solution sets every targeted α_i to either ℓ or u. On three small graphs (at most 8 nodes), try a grid of 21 values for one node's α and show that f is maximised at an end of the interval. In the report, explain why this lets greedy consider only two values per candidate.
3. Implement greedy for maximisation: in each round, for every candidate v not in T and for both $\alpha_v = \ell$ and $\alpha_v = u$, recompute f from scratch and keep the best. Run it on the 200 highest-degree nodes only, for k up to 10.

4. For each profile and each k among 1, 2, 3, 4, 5 and 10, the node added (original ID), the value of α it was given, f, the number of equilibrium computations, and the cumulative wall-clock time.

Task 2: Reproduce Abebe et al. (2018)

Implement the following from the paper. Work from the paper itself, and in the report note every place where you had to make a choice the paper does not specify.

1. **Fast greedy.** The paper's greedy heuristic, using rank-one (Sherman–Morrison) updates so that each candidate is evaluated without a new solve. Derive the update in the report. Run it on all nodes for k up to 30, for maximisation under both profiles and for minimisation under (U).
2. **Unbudgeted problem.** After substituting $X = A^{-1}$, the unbudgeted problem is convex. Solve unbudgeted maximisation and minimisation on ego-Facebook with projected gradient descent (derive the gradient yourself) from three starting points, and show that they reach the same value. On two small graphs (at most 15 nodes), check the answer by brute force over all 2^n extreme assignments, or with `cvxpy`.
3. **Baseline I** and **Baseline II**, as described in the paper.
4. **Non-submodularity.** Reproduce Example 13 of the paper numerically. Then find a pair $A \subset B$ and a node v on ego-Facebook where the marginal gain of v is larger for B than for A.

Check against Task 1– On the 200-node shortlist, fast greedy must reproduce your Task 1 results up to ties. Show this for k up to 10, and explain every mismatch.

Baselines– Include *Random* (mean of 20 random k-sets, each node given its better extreme value) in every comparison.

For each one of the following claims, say whether it holds on ego-Facebook and back the answer with your numbers:

1. The unbudgeted problem can be solved to optimality in polynomial time.
2. Some optimal solution uses only the extreme values ℓ and u. Check whether your unbudgeted solution does.
3. The budgeted objective is neither submodular nor supermodular.
4. Greedy does better than Baselines I and II at every budget.

Task 3: Exact reference by exhaustive search

The budgeted objective is not submodular, so greedy has no approximation guarantee. A brute-force optimum on a small pool shows what it actually loses.

1. Build the pool C as the union of the 15 highest-degree nodes and the 15 nodes with the best single-node gain (taking the better of ℓ and u for each). Do this for maximisation under each profile and report $|C|$ and how much the two lists overlap.
2. For $k_{gt} = 2$ and 3, try every subset of C of size k_{gt} and every assignment of ℓ or u to its members. Call the best one T^*.
3. Report T^* (original IDs with the assigned values), $f(T^*)$, the number of configurations evaluated, and the running time.

With $|C| = 30$ and $k_{gt} = 3$ there are 4,060 subsets with 8 assignments each. Inverting the matrix again for every configuration is too slow, so think about what you can reuse.

Task 4: Results, analysis and improvement

1. **Comparison** of greedy, Baseline I, Baseline II and Random:

   a. plot f against k (k = 1, 2, 3, 4, 5, 10, 20 and 30) for maximisation under (U) and (P) and for minimisation under (U), with the unbudgeted optimum and the value without intervention as horizontal lines;
   b. plot running time against k for the greedy of Task 1 and the fast greedy of Task 2;
   c. give one table with f at k = 5, 10 and 30, the fraction of the unbudgeted gain reached, total time and peak memory (use necessary library functions for this).
2. For k = 2 and 3, report the gain of greedy and of each baseline as a fraction of the gain of T^*, and the Jaccard overlap with T^*. Also run greedy restricted to C.
3. For the greedy targets, plot s_i against the average opinion of the neighbours of i, and mark whether each target was set to ℓ or u. Explain the pattern. Count how often, along the greedy run, a marginal gain was larger than the one in the previous round.
4. **[Bonus] One improvement on the paper and justify it:**
   a. Repeatedly swap a chosen node for one from the 200-node shortlist, or flip its value between ℓ and u, while f improves. Report how much this adds and how often greedy was already locally optimal. Explain why.

For the improvement, show its effect with numbers and a plot, and explain why it helps. If it does not help, explain why not.

EVALUATION GUIDELINES [TOTAL = 58]

1. Task 1: equilibrium and baseline greedy **[13 marks]**
   a. Equilibrium computation, the hand derivation and the extreme-value check: **8**
   b. Baseline greedy on the shortlist and the reported metrics: **5**
2. Task 2: reproduction of the paper **[17 marks]**
   a. Fast greedy with the derived rank-one update, matching Task 1: **8**
   b. Unbudgeted convex solution with the small-graph checks: **5**
   c. Baselines, the non-submodularity example and a verdict on each of the four claims: **4**
3. Task 3: exact reference by exhaustive search **[8 marks]**
4. Task 4: analysis and improvement **[15 marks]**
   a. Comparison table and plots, reference ratios and the study of the greedy choices: **10**
   b. **[Bonus]** A working improvement with evidence and an explanation: **+5 (will be added if score is less than the full marks)**
5. Report clarity, reproducibility and references **[5 marks]**
6. **Deductions:**
   a. Plagiarism (60% or more overlap) or a majority AI-generated flag: **100% penalty (-58 marks)**
   b. Using a library that does the algorithm for you: **-30 marks**
   c. Not following the naming convention: **-2 marks per violation**